STAT 410
Spring 2019
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Name ANSWERS

Version A

NetID _____

Exam 2

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Be sure to show all your work; your partial credit might depend on it.

Put your final answers at the end of your work, and mark them clearly.

No credit will be given without supporting work.

The exam is closed book and closed notes.
You are allowed to use a calculator and one 8½" x 11" sheet with notes.

Turn in all scratch paper with your exam.

Academic Integrity

The University statement on your obligation to maintain academic integrity is:

If you engage in an act of academic dishonesty, you become liable to severe disciplinary action. Such acts include cheating; falsification or invention of information or citation in an academic endeavor; helping or attempting to help others commit academic infractions; plagiarism; offering bribes, favors, or threats; academic interference; computer related infractions; and failure to comply with research regulations.

Article 1, Part 4 of the Student Code gives complete details of rules governing academic integrity for all students. You are responsible for knowing and abiding by these rules.

1. Let $\delta > 0$ and let $X_1, X_2, \ldots, X_n$ be a random sample of size n from the distribution with probability density function

$$f(x;\delta) = \frac{\delta^5 (\ln x)^4}{24\, x^{\delta+1}}, \qquad x > 1, \qquad \text{zero otherwise.}$$

a) (8) Show that $W = \ln X$ has a Gamma distribution. What are its parameters?

$$w = \ln x \qquad x = e^w \qquad \frac{dx}{dw} = e^w$$

$$f_W(w) = \frac{\delta^5 w^4}{24 \left(e^w\right)^{\delta+1}} \cdot e^w = \frac{\delta^5}{24} w^4 e^{-\delta w}$$

$$= \frac{\delta^5}{\Gamma(5)} w^{5-1} e^{-\delta w}, \qquad w > 0.$$

$\Rightarrow$ W has a $\text{Gamma}(\alpha = \mathbf{5}, \theta = \mathbf{\frac{1}{\delta}})$ distribution. $(\lambda = \delta)$

b) (6) Suppose $n = 4$ and $\delta = 4.5$. Find $P\left(\sum_{i=1}^{4} \ln X_i > 3\right)$.

$\sum_{i=1}^{n} W_i = \sum_{i=1}^{n} \ln X_i$ has a $\text{Gamma}(\alpha = 5n, \theta = \frac{1}{\delta})$ distribution. $(\lambda = \delta)$

$$P\left(\sum_{i=1}^{4} \ln X_i > 3\right) = P\left(\text{Gamma}(\alpha = 20, \theta = \frac{1}{4.5}) > 3\right)$$

$$= P(\text{Poisson}(4.5 \cdot 3) \le 20 - 1) = P(\text{Poisson}(13.5) \le 19) = \mathbf{0.942}.$$

$$P\left(\text{Gamma}(\alpha = 20, \theta = \frac{1}{4.5}) > 3\right) = P(\chi^2(40) > 27) \approx 0.942128.$$

1. (continued) $$f(x;\delta) = \frac{\delta^5 (\ln x)^4}{24\, x^{\delta+1}}, \qquad x > 1.$$

c) (4) Find a sufficient statistic $Y = u(X_1, X_2, \ldots, X_n)$ for δ.

$$f(x_1;\delta)\, f(x_2;\delta) \ldots f(x_n;\delta) = \prod_{i=1}^{n} \frac{\delta^5 (\ln x_i)^4}{24\, x_i^{\delta+1}}$$

$$= \frac{\delta^{5n}}{24^n} \cdot \left(\prod_{i=1}^{n} x_i \right)^{-\delta-1} \cdot \left(\prod_{i=1}^{n} \ln x_i \right)^4 .$$

$\Rightarrow$ $Y_1 = \prod_{i=1}^{n} X_i$ is a sufficient statistic for δ.

$\Rightarrow$ $Y_2 = \ln \prod_{i=1}^{n} X_i = \sum_{i=1}^{n} \ln X_i$ is also a sufficient statistic for δ.

OR

$$f(x;\delta) = \frac{\delta^5 (\ln x)^4}{24\, x^{\delta+1}} = \exp\{ -\delta \ln x + 5 \ln \delta - \ln 24 + 4 \ln \ln x - \ln x \}$$

$\Rightarrow$ $K(x) = \ln x.$

$\Rightarrow$ $Y_2 = \sum_{i=1}^{n} \ln X_i$ is a sufficient statistic for δ.

$\Rightarrow$ $Y_1 = e^{Y_2} = \prod_{i=1}^{n} X_i$ is also a sufficient statistic for δ.

1. (continued) $$f(x;\delta) = \frac{\delta^5 \left(\ln x\right)^4}{24\, x^{\delta+1}}, \qquad x > 1.$$

d) (10) Find the maximum likelihood estimator $\hat{\delta}$ of δ.

Suppose $n = 4$, and $x_1 = 1.03$, $x_2 = 1.3$, $x_3 = 3$, $x_4 = 5$.

Find the maximum likelihood estimate of δ.

$$L(\delta) = \prod_{i=1}^{n} \frac{\delta^5 \left(\ln x_i\right)^4}{24\, x_i^{\delta+1}} = \frac{\delta^{5n}}{24^n} \cdot \left(\prod_{i=1}^{n} x_i\right)^{-\delta-1} \cdot \left(\prod_{i=1}^{n} \ln x_i\right)^4.$$

$$\ln L(\delta) = 5n \ln \delta - n \ln 24 - \left(\delta+1\right) \cdot \sum_{i=1}^{n} \ln x_i + 4 \sum_{i=1}^{n} \ln \ln x_i.$$

$$\frac{d}{d\delta} \ln L(\delta) = \frac{5n}{\delta} - \sum_{i=1}^{n} \ln x_i = 0.$$

$$\Rightarrow \qquad \hat{\delta} = \frac{5n}{\sum_{i=1}^{n} \ln X_i}.$$

$n = 4$ $\qquad\qquad$ $\ln 1.03 + \ln 1.3 + \ln 3 + \ln 5 \approx 3.$

$$\hat{\delta} = \frac{5 \cdot 4}{3} = \mathbf{\frac{20}{3}} \approx 6.66667.$$

1. (continued) $f(x;\delta) = \dfrac{\delta^5 (\ln x)^4}{24\, x^{\delta+1}}, \qquad x > 1.$

e) (11) Suppose $n = 4,$ and $x_1 = 1.03,\ x_2 = 1.3,\ x_3 = 3,\ x_4 = 5.$

Construct a 95% confidence interval for δ.

"Hint": $\sum_{i=1}^{n} W_i = \sum_{i=1}^{n} \ln X_i$.

$Y = \sum_{i=1}^{n} W_i = \sum_{i=1}^{n} \ln X_i$ has a $\text{Gamma}(\alpha = 5n,\ \theta = \frac{1}{\delta})$ distribution. $(\lambda = \delta)$

$\Rightarrow \quad 2\delta \sum_{i=1}^{n} \ln X_i$ has a $\chi^2(2\alpha = 10n)$ distribution.

$\Rightarrow \quad P\left(\chi^2_{1-\alpha/2}(10n) < 2\delta \sum_{i=1}^{n} \ln X_i < \chi^2_{\alpha/2}(10n) \right) = 1 - \alpha.$

$\Rightarrow \quad P\left(\dfrac{\chi^2_{1-\alpha/2}(10n)}{2\sum_{i=1}^{n} \ln X_i} < \delta < \dfrac{\chi^2_{\alpha/2}(10n)}{2\sum_{i=1}^{n} \ln X_i} \right) = 1 - \alpha.$

A $(1-\alpha)100\,\%$ confidence interval for δ

$$\left(\frac{\chi^2_{1-\alpha/2}(10n)}{2\sum_{i=1}^{n} \ln X_i},\ \frac{\chi^2_{\alpha/2}(10n)}{2\sum_{i=1}^{n} \ln X_i} \right).$$

$n = 4 \qquad \ln 1.03 + \ln 1.3 + \ln 3 + \ln 5 \approx 3.$

$\chi^2_{0.975}(40) = 24.43, \qquad \chi^2_{0.025}(40) = 59.34.$

$\left(\dfrac{24.43}{2\cdot 3}, \dfrac{59.34}{2\cdot 3} \right)$ **(4.07, 9.89)**

2. Let $\beta > 0$ and let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a distribution with probability density function

$$f(x;\beta) = \frac{6x^5}{\beta^6}, \qquad 0 < x < \beta, \qquad \text{zero elsewhere.}$$

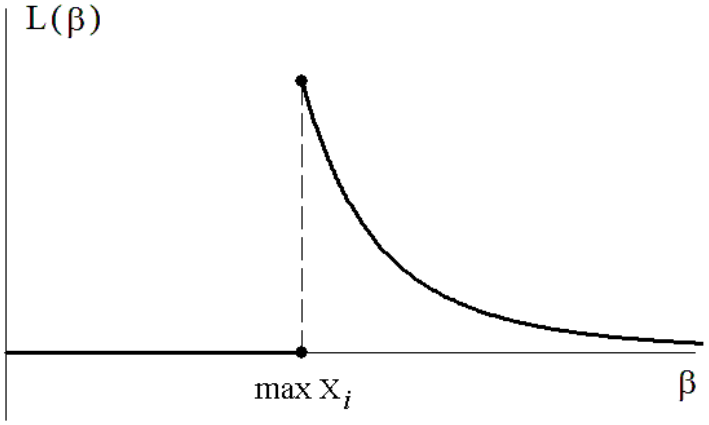
a) (10) (i) Find the maximum likelihood estimator for β, $\hat{\beta}$.

(ii) Suppose $n = 4$, and $x_1 = 1.5$, $x_2 = 2.3$, $x_3 = 2.9$, $x_4 = 3.3$. Find the maximum likelihood estimate for β, $\hat{\beta}$.

$$L(\beta) = \prod_{i=1}^{n}\left(\frac{6x_i^5}{\beta^6}\right) = \frac{6^n}{\beta^{6n}}\cdot\prod_{i=1}^{n} x_i^5, \qquad \beta > \max x_i,$$

$$L(\beta) = 0, \qquad \beta < \max x_i.$$

$$\ln L(\beta) = n\ln 6 - 6n\ln\beta + 5\sum_{i=1}^{n}\ln x_i. \qquad \frac{d}{d\beta}\ln L(\beta) = -\frac{6n}{\beta} = 0 \quad ???$$



The goal is to find where the maximum of $L(\beta)$ occurs.

Therefore,

$$\hat{\beta} = \mathbf{max\ X_i}.$$

$x_1 = 1.5$, $x_2 = 2.3$, $x_3 = 2.9$, $x_4 = 3.3$. $\Rightarrow$ $\hat{\beta} = \max x_i = \mathbf{3.3}$.

2. (continued) $f(x;\beta) = \dfrac{6x^5}{\beta^6}, \qquad 0 < x < \beta.$

b) (9) Is the maximum likelihood estimator $\hat{\beta}$ an unbiased estimator of β?

If $\hat{\beta}$ is not an unbiased estimator of β, construct an unbiased estimator of β based on $\hat{\beta}$.

Circle one: Yes | No

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\,dy = \int_0^x \frac{6y^5}{\beta^6}\,dy = \frac{x^6}{\beta^6}, \qquad 0 < x < \beta.$$

$$F_{\max X_i}(x) = \left[F_X(x)\right]^n = \left[\frac{x^6}{\beta^6}\right]^n = \left[\frac{x}{\beta}\right]^{6n} = \frac{x^{6n}}{\beta^{6n}}, \qquad 0 < x < \beta.$$

$$f_{\max X_i}(x) = F'_{\max X_i}(x) = \frac{6n \cdot x^{6n-1}}{\beta^{6n}}, \qquad 0 < x < \beta.$$

$$E\left(\hat{\beta}\right) = E\left(\max X_i\right) = \int_{-\infty}^{\infty} \left(x \cdot f_{\max X_i}(x)\right) dx = \int_0^{\beta} \left(x \cdot \frac{6n \cdot x^{6n-1}}{\beta^{6n}}\right) dx$$

$$= \frac{6n}{\beta^{6n}} \cdot \int_0^{\beta} x^{6n}\,dx = \frac{6n}{6n+1} \cdot \beta \neq \beta.$$

$\Rightarrow$ $\hat{\beta}$ is NOT an unbiased estimator of β.

$$E\left(\frac{6n+1}{6n}\,\hat{\beta}\right) = \beta.$$

$\Rightarrow$ $\dfrac{6n+1}{6n}\,\hat{\beta} = \dfrac{6n+1}{6n}\max X_i$ is an unbiased estimator of β.

2. (continued) $f(x;\beta) = \frac{6x^5}{\beta^6}, \qquad 0 < x < \beta.$

c) (8) Construct a consistent estimator of β based on $\sum_{i=1}^{n} X_i^3$.

By WLLN, $\frac{1}{n}\sum_{i=1}^{n} X_i^3 \xrightarrow{P} E(X^3) = \int_0^{\beta} x^3 \cdot \frac{6x^5}{\beta^6}\,dx = \frac{6}{9}\cdot\beta^3 = \frac{2}{3}\cdot\beta^3.$

Consider $g(x) = \sqrt[3]{1.5x} = \sqrt[3]{\frac{3}{2}x}$.

$g(\frac{2}{3}\cdot\beta^3) = \beta.$ $g(x)$ is continuous at $\frac{2}{3}\cdot\beta^3$.

$\Rightarrow \quad g(\frac{1}{n}\sum_{i=1}^{n} X_i^3) = \sqrt[3]{\frac{1.5}{n}\sum_{i=1}^{n} X_i^3} \xrightarrow{P} g(\frac{2}{3}\cdot\beta^3) = \beta.$

$\tilde{\tilde{\beta}} = \sqrt[3]{\frac{1.5}{n}\sum_{i=1}^{n} X_i^3}$ is a consistent estimator of β.

3. (9) Suppose the GPA (X) and the starting salary $(Y$, in thousands of dollars$)$ for the Philosophical Engineering majors who graduated from Anytown State University follow a Bivariate Normal distribution with

$$\mu_X = 3.3, \quad \sigma_X = 0.2, \quad \mu_Y = 73, \quad \sigma_Y = 7, \quad \rho = 0.6.$$

Find the probability that a student who graduated with the GPA of 3.7 would have the starting salary over \$80,000. That is, find $P(Y > 80 \mid X = 3.7)$.

Given $X = 3.7$, Y has Normal distribution

with mean $\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X) = 73 + 0.6 \cdot \frac{7}{0.2} \cdot (3.7 - 3.3) = 81.4$

and variance $\left(1 - \rho^2\right) \cdot \sigma_Y^2 = \left(1 - 0.6^2\right) \cdot 7^2 = 31.36.$

$$P(Y > 80 \mid X = 3.7) = P\left(Z > \frac{80 - 81.4}{\sqrt{31.36}}\right) = P(Z > -0.25) = \mathbf{0.5987}.$$